

The Ghost in the Machine

PRECOG TASK – Natural language precessing

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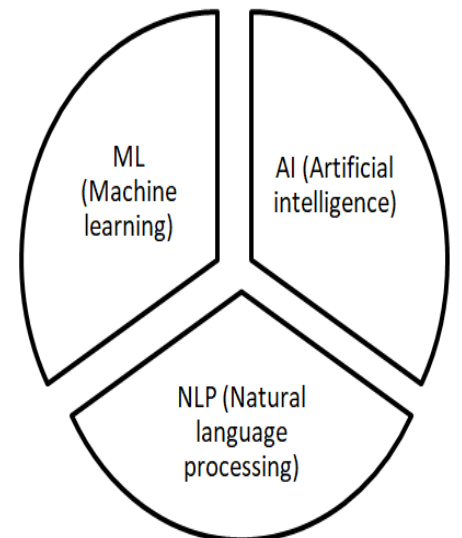
TASK-0

Goal

- Construct a text dataset where **authorship** is the dominant variable
- Control for **topic**, so stylistic signals are isolated
- Create human-written vs AI-generated corpora under matched conditions
- **Key Idea**
- If topic is held constant, differences in text should reflect *style*, not content.

Three Classes

- **Class 1:** Human-written literary text (Project Gutenberg)
- **Class 2:** AI-generated text (neutral style)
- **Class 3:** AI-generated text (author-mimicking style)
- **Why three classes?**
- Compare **Human vs AI**
- Compare **AI vs AI-style-transfer**
- Test whether “style imitation” closes the human-AI gap



Human Text (Class 1)

- Removed boilerplate, headers, and metadata
- Reconstructed paragraph-like units by grouping sentences
- Preserves vocabulary, syntax, and rhythm despite lost formatting

AI Text (Classes 2 & 3)

- Gemini API used for controlled generation
- 500 paragraphs per class
- Paragraph length: ~100–200 words

What Is Controlled

- | | | |
|----------------------|------------------|--------------|
| • Topic distribution | Paragraph length | Dataset size |
|----------------------|------------------|--------------|

What Is Allowed to Vary

- | | |
|---------------------|----------------------|
| • Lexical choice | Syntactic complexity |
| • Punctuation usage | Sentence rhythm |

• **Why This Is Important**

- Any detectable differences arise from *style*, not subject matter.

TASK-0

Outcome & Purpose

- **Final Dataset**
- ~1500 total paragraphs
- Balanced across authorship conditions
- Clean, comparable text units
- **Enables**
- Quantitative stylistic analysis (Task 1)
- Multi-tier AI detection (Task 2)
- Interpretability & error analysis (Task 3)
- **Core Insight**
- Style leaves a measurable fingerprint — even when content is identical.

TASK-1

- Objective
- Before training classifiers, verify that **Human** and **AI-generated** texts are *mathematically distinguishable*.
- Focus on **stylistic signals**, not semantic content.
- Key Observations & Takeaways
- Observed Patterns
- AI-Neutral text showed:
 - 1) Lower lexical diversity
 - 2) More uniform sentence structure
- AI-Mimic partially matched Dickens:
 - 1) Increased dependency depth
 - 2) Higher punctuation density
- Human text retained:
 - 1) Irregular vocabulary usage
 - 2) More natural syntactic variance
- Conclusion
- Even when topics are identical, **authorship leaves a measurable stylistic fingerprint**.
- These features justify using statistical and ML-based detectors in Task 2.

TASK-1

- **Why This Matters**
- If models succeed later, it's because the signal already exists — not because of overfitting.

TASK-2

The statistician-

Purpose

- Test if **simple surface statistics alone** can separate AI from Human text

The Semanticist

Purpose

- Capture **semantic smoothness and coherence**
- Detect distributional differences invisible to raw statistics
- Slide 3 — Tier C, Results & Interpretation

Tier C: The Transformer

Purpose

- Learn **deep stylistic and structural cues**
- Serve as the strongest detector

TASK-2

Key Observations

- Tier A works when AI text is stylistically uniform
- Tier B improves when meaning distribution differs
- Tier C performs best, but still makes **human-like mistakes**

Negative Results Matter

- Near-50% accuracy in some settings suggests:
 - Strong AI imitation
 - Overlapping stylistic space
 - Limits of authorship detection

Conclusion

- If AI can fool deeper models, the boundary between human and machine writing is no longer purely stylistic.

TASK-3

Objective

- Understand *why* the Transformer (Tier C) labels text as **AI-generated**.
- Move beyond accuracy into **model interpretability**.

Approach

- Use **SHAP (Shapley Additive Explanations)** on the fine-tuned DistilBERT + LoRA model.
- Analyze **token-level contributions** to the AI vs Human decision.

What SHAP Provides

- Highlights words and phrases that:
 - Push predictions toward **AI**
 - Push predictions toward **Human**
- Allows inspection of both **vocabulary choice** and **sentence rhythm**.

TASK 3

- **Saliency Mapping: What the Model Notices**
- **Key Observations from SHAP**
- The model strongly reacts to:
 - Polished, abstract vocabulary (e.g., “*intricate*,” “*testament*,” “*delve*”)
 - Balanced, well-structured clauses
 - Formal discourse markers and transitions
- **Important Insight**
- The model is **not only keyword-based**
- Saliency often spans:
 - Long sentence arcs
 - Clause-level structure
 - Rhythm and symmetry in prose
- **Interpretation**
- The detector learns a *stylistic smoothness prior* rather than explicit topic cues.

TASK 3

Error Analysis & Implications

False Positives (Human → AI)

- Occurred when:
 - Human text was highly formal or academic
 - Repetitive sentence structures appeared
 - Prose lacked idiosyncratic variation

False Negatives (AI → Human)

- Occurred when:
 - AI text introduced irregular phrasing
 - Sentence rhythm mimicked human inconsistency
 - Vocabulary avoided common “AI-isms”

Conclusion

- The boundary between Human and AI writing is **stylistic, not absolute**
- High-quality AI can cross into the human stylistic manifold

TASK 4

Objective

- Test whether AI-generated text can be **iteratively optimized** to bypass the strongest detector (Tier C).
- Treat the detector as a *black box adversary*.
- **Core Question**
- If the model relies on stylistic cues, can those cues be systematically removed?

Key Concept

- Frame AI detection as an **optimization problem**:
 - Input: AI-generated paragraph
 - Output: Probability of being classified as *Human*

TASK-4

- **Genetic Algorithm Workflow**
- **Initial Population**
- Generate 10 AI “Imposter” paragraphs using Gemini
- All paragraphs on the same topic
- **Fitness Function**
- Human-confidence score from Tier C Transformer
- **Selection**
- Keep top-3 paragraphs with highest Human probability
- **Mutation (LLM-as-Mutator)**
- Gemini rewrites selected paragraphs using targeted prompts:
 - Change sentence rhythm
 - Introduce slight grammatical irregularities
 - Insert rare or archaic words

TASK-4

- **Observed Outcome**
- Human confidence score **increases across generations**
- Final AI paragraphs often exceed **90% “Human” probability**
- **Why This Works**
- Detector relies on:
 - Smoothness
 - Regular sentence structure
 - Polished transitions
- Mutations disrupt these signals without harming semantics
- **Final Insight**
- AI detection is an *arms race*, not a fixed boundary.
- **Personal Test**
- Human essays can be misclassified as AI
- AI text can be evolved to appear convincingly human



FINAL CONCLUSIONS



- **Final Conclusions**
- **Authorship leaves a measurable fingerprint.**
Even when topics are controlled, human and AI writing differ in lexical diversity, syntactic structure, punctuation usage, and readability.
- **Surface statistics capture part of the signal, but not all of it.**
Simple features (Task 1, Tier A) can distinguish AI text when it is stylistically uniform, but they fail as AI becomes more expressive.
- **Semantic and transformer-based models perform better, but are not infallible.**
Tier B and Tier C models show improved accuracy, yet still confuse highly formal human writing with advanced AI-generated prose.
- **Detection models rely on stylistic regularities rather than true “intent.”**
Saliency analysis reveals sensitivity to polished vocabulary, balanced sentence rhythm, and structural smoothness—patterns that AI can learn to imitate or avoid.
- **AI detection is adversarial, not absolute.**
Through iterative mutation (Task 4), AI-generated text can be evolved to bypass even strong detectors, reaching high “Human” confidence.